$25.14-$25.10 and a 100 share bid at $25.09, the five bids of relatively insignificant notional value at $25.14-$25.10 would prevent the bid of relatively significant notional value at $25.09 from being captured by the definition of depth of book data. For this reason, the proposed definition of depth of book data included a "minimum size requirement" for depth price levels—namely, the presence of a "bid or offer," which incorporates the concept of a "round lot."[427]

That said, in response to the commenter's request, the Commission is clarifying that the requirement for the presence of a bid or offer in determining the five price levels does not require a "pure" or "unitary" round lot consisting of only one order to be present at a price level. Rather, odd-lots that aggregate into a round lot pursuant to the prescribed method could also establish a price level, as long as there is at least one round lot of interest—unitary or aggregated—on at least one SRO. As a round lot reflects trading interest of meaningful size to market participants, the Commission believes that odd-lots that in the aggregate reflect size equivalent to a round lot also represent trading interest of meaningful size and should be included in depth of book data. The following example illustrates how odd-lot aggregation would operate in the depth of book context.

Example 2: Odd-Lot Aggregation at Depth

| | Number of Shares Bid for Stock X | | |
|---|---|---|---|
| Price | Exchange A | Exchange B | Exchange C |
| $25.50 | 100 (NBB) | 0 | 0 |
| $25.49 | 0 | 100 | 0 |
| $25.48 | 0 | 0 | 50 |
| $25.47 | 60 | 0 | 50 |

[427] Proposing Release, 85 FR at 16754.

| | | | |
|---|---|---|---|
| $25.46 | 0 | 0 | 0 |
| $25.45 | 60 | 0 | 0 |
| $25.44 | 0 | 25 | 0 |
| $25.43 | 0 | 75 | 0 |
| $25.42 | 100 | 0 | 0 |
| $25.41 | 0 | 0 | 2 |
| $25.40 | 0 | 5 | 0 |
| $25.39 | 0 | 0 | 100 |

Here, the first five prices at which there is a bid that is lower than the NBB—measured in terms of the presence of at least one singular or aggregated round lot on at least one SRO—are $25.49 (100 shares on Exchange B), $25.47 (50 shares at $25.48 and 50 shares at $25.47 on Exchange C, aggregated and displayed at the less aggressive price), $25.45 (60 shares at $25.47 and 60 shares at $25.45 on Exchange A, aggregated and displayed at the less aggressive price), $25.43 (25 shares at $25.44 and 75 shares at $25.43 on Exchange B, aggregated and displayed at the less aggressive price), and $25.42 (100 shares on Exchange A). Hence, in Example 2, depth of book data would include: 100 shares at $25.49, $25.47, $25.45, $25.43, and $25.42 (with attribution to the relevant SRO, as explained above).

One commenter argued that allowing aggregation across multiple price levels to determine whether there is a round lot bid or offer at a particular depth of book price level could create significant computational issues, particularly when orders are cancelled and the five price levels would have to be redetermined, possibly resulting in diminished competing consolidator performance and higher capacity requirements for downstream users of the data. This

commenter also argued that displaying depth of book price levels in this manner could potentially lead to a deceptive view of market activity at prices that are nowhere near current market prices, raising concerning issues related to investor protection.[428]

The Commission acknowledges that aggregating odd-lots to determine depth of book price levels will require computation by competing consolidators and that new orders or order cancelations and modifications could require the five levels to be redetermined. However, alternatives such as the inclusion of all odd-lots at inferior prices would require more outbound message traffic from competing consolidators to subscribers and would therefore raise similar concerns about system performance and the capability of subscribers to consume the data.[429] In addition, as explained below and in the Proposing Release, the decentralized consolidation model will foster a competitive environment for the provision of consolidated market data, and the Commission believes that this will lead to improvements in the use of more competitive, low-latency aggregation and transmission technologies, mitigating concerns about computational and performance issues related to the generation and dissemination of depth of book data.[430] Competing consolidator subscribers also have the option of consuming varying levels of depth of book data, depending on their needs, and could determine that the latency costs of consuming several levels of depth exceed the benefits. Furthermore, the Commission does not believe that aggregating odd-lots across prices will lead to a deceptive view of market activity or mislead investors because exchanges currently aggregate odd-lots across price levels to form round lots

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428 CBOE Letter at 19.

429 As discussed above, other alternatives presented by commenters, such as including five price levels of depth of book information without regard to whether those price levels are round lots, have other drawbacks. See supra notes 423–427 and accompanying text.

430 See Proposing Release, 85 FR at 16768; infra Section III.B.5.

and provide their best bids and offers to the exclusive SIPs at the least aggressive price of all such odd-lots. In addition, Commission rules, which will include this methodology, will provide transparency to market participants regarding the inclusion in core data of odd-lots at depth of book price levels aggregated pursuant to this methodology, which addresses the concern that cross-price odd-lot aggregation would lead to a deceptive view of market activity or mislead investors.

**(f) Insufficient Justification for Inclusion of Depth of Book Data in Core Data**

Some commenters questioned whether there is sufficient need among market participants for depth of book data to be included in core data.[431] One commenter stated that depth of book data is for "serious traders" and is already available on a proprietary basis for anyone who needs it, and that it is not clear whether there would be demand for the "truncated" set of information reflected in the proposed definition of depth of book from any particular set of market participants.[432] In a subsequent letter, this commenter stated that the Commission's depth of book data proposal "lacks data driven analysis" and that the Commission "released no independent studies or research analyzing the impact of this proposal or addressing the five price level demarcation."[433] Similarly, another commenter stated that depth of book data is not "necessary or helpful for all investors" and that its inclusion in core data would increase core

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431 See Nasdaq Letter IV at 33; TD Ameritrade Letter at 5; NYSE Letter II at 3–4.

432 See Nasdaq Letter IV at 33.

433 See Nasdaq Letter V at 5.

data costs unnecessarily and create confusion as to what is necessary for regulatory compliance.[434]

The Commission disagrees with these comments. As evidenced by the support for including five levels of depth of book data expressed by a wide range of commenters—including exchange, buy-side, and sell-side market participants, many of whom explained with specificity why five levels of depth of book would meet their needs[435]—the Commission believes there is demand for core data that includes depth of book data, including the definition of depth of book data, as modified and adopted herein. The Commission also believes that including depth of book data in core data would promote a broader distribution of this data among market participants, particularly those who do not currently subscribe to the proprietary depth of book products offered by the exchanges.[436] Nevertheless, the Commission acknowledges that depth of book data may not be "necessary or helpful for all investors"[437] in a direct sense, but even investors who do not directly consume depth of book data will benefit from it indirectly,[438] since many of their broker-dealer intermediaries would likely use depth of book data for improved order placement,[439] which ultimately will improve the execution quality of customer orders.[440]

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434 See TD Ameritrade Letter at 5.

435 See supra note 388.

436 See infra note 1974 and accompanying text.

437 See TD Ameritrade Letter at 5.

438 See supra Section II.C.2(a).

439 See infra note 879 and accompanying text.

440 These indirect benefits would accrue to customer orders executed by broker-dealers that do not currently use proprietary depth of book data but that would use the depth of book data as adopted and as included in core data.

Furthermore, the Commission has explained the implications of these amendments on best execution obligations in detail.[441] Therefore, the inclusion of depth of book data in core data should not "cause confusion as to what may be required for regulatory compliance."[442] Finally, the Commission disagrees that the definition of depth of book data is unsupported by "data driven analysis" because the Commission considered staff analyses of depth of book data in both proposing[443] and adopting[444] the definition.

**(g) Need for Competing Consolidators to Offer Depth of Book**

Some commenters supported requiring SROs to provide depth of book information to competing consolidators but opposed requiring competing consolidators to include depth of book in the products offered to their subscribers[445] or emphasized that competing consolidators should be provided with enough raw data to be competitive with proprietary offerings but permitted to offer a range of products due to investors' diverse market data needs.[446]

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441 See supra Section I.E (discussing the implications of these amendments generally for best execution obligations); infra Section II.F.2(h) (discussing best execution obligations in the context of depth of book data).

442 See TD Ameritrade Letter at 5.

443 See Proposing Release, 85 FR at 16754.

444 See supra note 387.

445 See BestEx Research Letter at 2, 5 ("Given that exchanges have zero competition in providing their own market data feeds, we welcome the Commission's proposed mandate that exchanges provide depth of book information and auction information to competing SIP vendors, thus reducing information asymmetry among market participants. However, we believe that SIP providers should not be required to include that information in their products.").

446 See BlackRock Letter at 2–3.

The Commission agrees that competing consolidators should be permitted to offer customers a range of products, including customized depth of book products that include more or less depth of book information than set forth in the definition of depth of book data. Modifying the requirements of Rule 614 so that competing consolidators will only be required to generate and offer one or more consolidated market data products, which can contain some or all of the elements of consolidated market data, will enable competing consolidators to specialize in different products to address their subscribers' market data needs.[447] Competing consolidators that receive proprietary data products from SROs to create products that go beyond consolidated market data (e.g., full depth of book data), would compensate SROs for this use pursuant to individual SRO fee schedules, while competing consolidators that limit their use of SRO data to the creation of products that include consolidated market data or a subset thereof would be charged pursuant to the effective national market system plan(s) fee schedules. If the effective national market system plan(s) establishes fees for data content underlying consolidated market data offerings that use subsets of consolidated market data, the competing consolidator would have the option of providing customized products that do not, for example, include all five levels of depth of book data, including products providing only the NBBO and/or the top of book quotes of exchanges.[448]

**(h) Best Execution Obligations regarding Depth of Book Data**

Some commenters discussed market participants' duty of best execution in light of the inclusion of depth of book data in core data. Some commenters noted that including depth of

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447 See infra Sections III.C.1(b); III.C.8(a).

448 See infra Section III.E.2(e).

book data in core data would assist market participants in fulfilling their best execution requirements.[449] Others went further and stated that depth of book data is currently necessary to fulfill their best execution obligations.[450] However, one commenter stated that "[s]upplementing core market data with depth-of-book data would confound market participants in fulfilling their best execution obligations."[451]

As explained above, these amendments do not change the duty of best execution.[452] Rather, the availability of additional data content as core data—including depth of book data—may be relevant to a broker-dealer's ability to achieve and analyze best execution, and broker-dealers should consider the availability of this information in connection with their best execution obligations.[453] However, for the reasons stated above, the Commission is not setting forth minimum data elements needed to achieve best execution or specifying the data elements that may be relevant to any specific situation or customer.[454] In addition, the Commission has explained the implications of these amendments on best execution obligations in detail,[455] and

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449 See, e.g., RBC Letter at 4; SIFMA Letter at 3–4; T. Rowe Price Letter at 1; Clearpool Letter at 1, 11 (stating that data content currently available only in proprietary feeds is necessary for best execution); Capital Group Letter at 2; McKay Letter at 1; Better Markets Letter at 1–2; DOJ Letter at 4.

450 See State Street Letter at 2; IEX Letter at 5.

451 Nasdaq Letter IV at 7. See also FINRA Letter at 7 (stating generally that the proposed changes to the content of consolidated market data and the manner in which it would be disseminated raise questions regarding best execution requirements and that the Commission should consider providing best execution guidance for broker-dealers).

452 See supra Section I.E.

453 See id.

454 See id.

455 See id.

does not agree that market participants will be "confounded" in fulfilling their best execution obligations by the addition of depth of book data to core data.[456]

## G. Definition of "Auction Information" under Rule 600(b)(5)

### 1. Proposal

The Commission proposed to define "auction information" as an element of core data. Specifically, "auction information" would be defined as all information specified by national securities exchange rules or effective national market system plans that is generated by a national securities exchange leading up to and during an auction—including opening, reopening, and closing auctions—and disseminated during the time periods and at the time intervals provided in such rules and plans. Auctions have become increasingly important liquidity events in recent years and have come to represent a significant proportion of overall trading volume. The Commission proposed to include auction information in core data to promote more informed and effective trading in auctions, which could also facilitate price formation and improve execution quality for more traders and investors.[457] The Commission solicited comment on the proposed definition of auction information, including the scope of auction-related information that should be included in the definition.[458]

### 2. Final Rule and Response to Comments

Many commenters favored including auction information in core data, but some opposed doing so. The Commission is adopting the definition of auction information, as proposed, with

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456 See supra note 451 and accompanying text. See also supra note 92.

457 See Proposing Release, 85 FR at 16759.

458 See id.

one modification.[459] Specifically, the Commission is adding the word "publicly" before "disseminated" to specify that only auction information that is publicly disseminated on an exchange's proprietary feeds is included in the definition of auction information and hence as an element of core data. The Commission believes that this modification will help ensure that all auction information that an exchange includes in its proprietary feeds will be included in core data, addressing the information asymmetries that currently exist between users of SIP data and proprietary data and facilitating the ability of core data subscribers to participate in auctions in an informed and effective manner.[460] This modification would also clarify that auction information does not include auction-related information that is made available to a limited group of market participants under certain exchange models, but not made publicly available.[461]

**(a) Support for Inclusion of Auction Information in Core Data**

Many commenters supported the inclusion of auction information in core data, emphasizing the importance of this information in light of the increasing proportion of transaction volume that takes place during opening and closing auctions.[462] One commenter

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459 The adopted definition of auction information also includes one technical change from the proposed definition—using the plural "auctions" rather than the singular "an auction" in the phrase "generated by a national securities exchange leading up to and during auctions." The plural form is more consistent with the use of "auctions" in the next clause of the definition. See infra Section VIII.

460 See Proposing Release, 85 FR at 16758–59.

461 See NYSE Rule 123C(6)(b) (relating to the dissemination of certain auction-related information to floor brokers).

462 See IEX Letter at 6; MEMX Letter at 5–6; Cboe Letter at 21; BlackRock Letter at 2; Fidelity Letter at 5; Schwab Letter at 5; State Street Letter at 2–3; T. Rowe Price Letter at 2; Capital Group Letter at 2; ICI Letter at 9–10; SIFMA Letter at 7; Citadel Letter at 1 (supporting the inclusion of auction information in core data as long as it does not materially increase latency); Virtu Letter at 5; IntelligentCross Letter at 4; STANY Letter